

# Seminar 5: Sampling and Aliasing

## Questions and Answers

### 1. Sampling a continuous-time sinusoid

Consider the continuous-time sinusoid  $x(t) = 4\cos(2\pi 300t + \pi/6)$ .

(a) Sample the signal at  $f_s = 2000$  samples/s and write the resulting sequence  $x[n]$ .

**Answer:** Sampling at 2000 samples/s gives  $T_s = 1/2000$  s, so the continuous-time frequency 300 Hz becomes  $300/2000 = 0.15$  cycles/sample.

$$x[n] = 4 \cos(0.3\pi n + \pi/6).$$

(b) Determine the sampling period  $T_s$ .

**Answer:**

$$T_s = 1/f_s = 1/2000 = 0.0005 \text{ s} = 0.5 \text{ ms}.$$

(c) Determine the normalized cyclic frequency in cycles/sample and the normalized radian frequency in rad/sample.

**Answer:**

$$f_0 = f_0/f_s = 300/2000 = 0.15 \text{ cycles/sample},$$

$$\omega_0 = 2\pi f_0 = 0.3\pi \text{ rad/sample}.$$

(d) How many samples are taken during one period of the sinusoid?

**Answer:**

$$N_T = f_s/f_0 = 2000/300 = 20/3 \approx 6.67 \text{ samples/period}.$$

(e) Repeat the normalized-frequency calculation for  $f_s = 800$  samples/s. Is the sinusoid still sampled above the Nyquist rate?

**Answer:**

$$f_0 = 300/800 = 0.375,$$

$$\omega_0 = 0.75\pi \text{ rad/sample}.$$

The Nyquist frequency is 400 Hz, which is above 300 Hz, so the sinusoid is still sampled above the Nyquist rate.

### 2. Sampling a multi-sinusoidal signal

Consider  $x(t) = 2 + 3\cos(2\pi 400t - \pi/4) + 2\sin(2\pi 900t)$ .

(a) Sketch the two-sided spectrum of  $x(t)$ , including the DC component, and label all frequencies and complex amplitudes.

**Answer:** Write the sine as a cosine with phase  $-\pi/2$ .

$$X_0 = 2,$$

$$X_{\{+400\}} = 1.5e^{-j\pi/4}, \quad X_{\{-400\}} = 1.5e^{j\pi/4},$$

$$X_{\{+900\}} = e^{-j\pi/2} = -j, \quad X_{\{-900\}} = e^{j\pi/2} = j.$$

(b) Determine the highest frequency present in the signal.

**Answer:**

$$f_{\{max\}} = 900 \text{ Hz.}$$

(c) Determine the theoretical minimum sampling rate required by the sampling theorem.

**Answer:**

$$f_s > 2f_{\{max\}} = 1800 \text{ samples/s.}$$

(d) Would  $f_s = 2000$  samples/s be sufficient? Would  $f_s = 1600$  samples/s be sufficient? Explain.

**Answer:** 2000 samples/s: yes. 1600 samples/s: no (Nyquist frequency 800 Hz < 900 Hz).

(e) Propose a practical sampling rate that provides some oversampling margin.

**Answer:** Example:  $f_s = 2400$  samples/s.

### 3. Different analogue frequencies, identical samples

A discrete-time sinusoid is  $x[n] = \cos(0.4\pi n + \pi/5)$ .

(a) Assume  $f_s = 1000$  samples/s. Find one continuous-time sinusoid with frequency between 0 and 500 Hz that produces  $x[n]$ .

**Answer:**

$$f_0 = (0.4\pi/2\pi)f_s = 0.2(1000) = 200 \text{ Hz,}$$

$$x_c(t) = \cos(2\pi \cdot 200t + \pi/5).$$

(b) Find two additional positive analogue frequencies below 2000 Hz that generate the same sample values.

**Answer:** Examples are 1200 Hz with phase  $+\pi/5$  and 1800 Hz with phase  $-\pi/5$ .

(c) Find one folded alias and determine the phase required so that its samples are identical to  $x[n]$ .

**Answer:**

$$f_{\{fold\}} = f_s - f_0 = 800 \text{ Hz,}$$

$$\varphi_{\{fold\}} = -\pi/5.$$

The phase changes sign because folding changes the sign of the normalized frequency.

(d) Explain why the original analogue frequency cannot be determined uniquely from  $x[n]$  alone.

**Answer:** The original frequency is not unique because analogue aliases separated by  $f_s$ , including folded aliases, produce the same samples.

### 4. Families of aliases

A sinusoid  $x(t) = A\cos(2\pi f_0 t + \phi)$

is sampled at  $f_s$  samples/s.

(a) Write the family of analogue frequencies  $f = f_0 + kf_s$  that produce the same normalized frequency.

**Answer:**

$$f = f_0 + kf_s, \quad k \in \mathbb{Z},$$

with the same phase  $\phi$ .

(b) Using cosine symmetry, write the family of folded aliases.

**Answer:**

$$f = -f_0 + kf_s, \quad k \in \mathbb{Z},$$

or equivalently positive folded frequencies  $|kf_s - f_0|$ .

- (c) State what happens to the phase when a frequency is folded.

**Answer:**

$$\varphi \rightarrow -\varphi.$$

The phase reverses sign for the folded family.

- (d) For  $f_0 = 180$  Hz,  $f_s = 500$  Hz, and  $\phi = \pi/3$ , list all positive aliases below 1200 Hz and give the phase associated with each.

**Answer:** The positive aliases below 1200 Hz are:

$$180, 680, 1180 \text{ Hz with phase } +\pi/3,$$

$$320, 820 \text{ Hz with phase } -\pi/3.$$

## 5. Sampling theorem and oversampling

A measurement system records a signal known to contain no frequency components above 7.5 kHz.

- (a) According to the sampling theorem, what condition must the sampling rate satisfy for exact reconstruction?

**Answer:**

$$f_s > 2(7.5 \text{ kHz}) = 15 \text{ kHz}.$$

- (b) Is a sampling rate of 15 kHz sufficient in the strict statement of the theorem? Explain.

**Answer:** No. The strict condition is  $f_s > 15$  kHz.

- (c) Would 20 kHz be a valid choice? What is the Nyquist frequency in that case?

**Answer:**

$$f_N = f_s/2 = 10 \text{ kHz}.$$

Yes.

- (d) Explain why practical systems normally sample somewhat above the theoretical minimum.

**Answer:** To allow a realizable anti-aliasing filter transition band and practical margin.

- (e) If the signal is sampled at 48 kHz, calculate the oversampling factor relative to twice the highest signal frequency.

**Answer:**

$$OSR = 48/15 = 3.2.$$

## 6. Discrete-time frequency and the principal Nyquist interval

Consider the sampled sinusoid  $x[n] = \cos(1.6\pi n)$ .

- (a) Reduce the normalized radian frequency to an equivalent value in the principal interval  $-\pi \leq \omega_{\text{hat}} \leq \pi$ .

**Answer:**

$$1.6\pi - 2\pi = -0.4\pi.$$

Thus the principal-interval frequency is  $-0.4\pi$  rad/sample.

- (b) Write  $x[n]$  using that principal-interval frequency.

**Answer:**

$$x[n] = \cos(0.4\pi n).$$

- (c) Sketch the spectral line locations over the normalized-frequency range  $-3\pi \leq \omega_{\text{hat}} \leq 3\pi$ , showing the  $2\pi$  periodicity.

**Answer:** Within  $-3\pi \leq \omega \leq 3\pi$ , the spectral lines occur at.

$$\hat{\omega}/\pi = \pm 0.4 + 2k,$$

$$\hat{\omega}/\pi = -2.4, -1.6, -0.4, 0.4, 1.6, 2.4.$$

- (d) Explain why the discrete-time spectrum repeats every  $2\pi$ .

**Answer:**

$$e^{j(\hat{\omega} + 2\pi)n} = e^{j\hat{\omega}n}.$$

## 7. Undersampling and frequency folding

A sinusoid of frequency  $f_0 = 1000$  Hz is sampled at each of the following rates: 3000, 2200, 1800, 1250, and 800 samples/s.

- (a) For each sampling rate, determine whether the sampling theorem is satisfied.

**Answer:** Satisfied: 3000, 2200 samples/s. Undersampled: 1800, 1250, 800 samples/s.

- (b) For every undersampled case, determine the alias frequency that lies between 0 and  $f_s/2$ .

**Answer:**

$$f_s = 1800: f_a = |1000 - 1800| = 800 \text{ Hz},$$

$$f_s = 1250: f_a = |1000 - 1250| = 250 \text{ Hz},$$

$$f_s = 800: f_a = |1000 - 800| = 200 \text{ Hz}.$$

- (c) For each case, calculate the corresponding normalized radian frequency in the principal interval.

**Answer:**

$$3000: \hat{\omega} = 2\pi/3,$$

$$2200: \hat{\omega} = 10\pi/11,$$

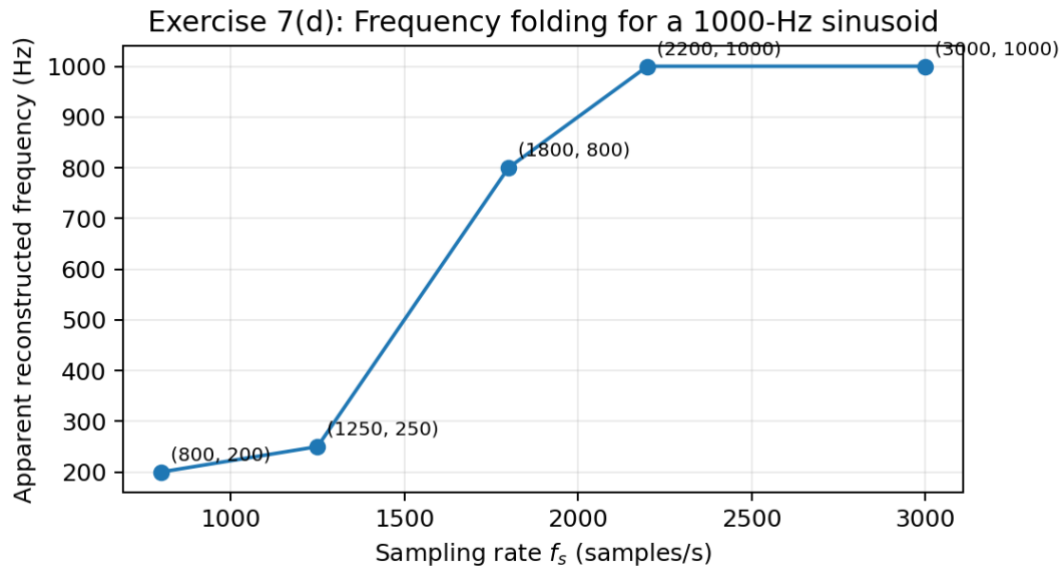
$$1800: \hat{\omega} = -8\pi/9 \text{ (cosine equivalent } + 8\pi/9),$$

$$1250: \hat{\omega} = -0.4\pi \text{ (cosine equivalent } + 0.4\pi),$$

$$800: \hat{\omega} = 0.5\pi.$$

- (d) Make a plot of apparent reconstructed frequency versus sampling rate and identify where folding occurs.

**Answer:** The apparent reconstructed frequencies for the five sampling rates are 1000, 1000, 800, 250, and 200 Hz, respectively.



*Apparent reconstructed frequency for the five sampling rates.*

## 8. Sampling followed by ideal reconstruction

Consider the system in which  $x(t)$  is sampled at  $f_s$  samples/s and then passed through an ideal discrete-to-continuous reconstruction stage, producing  $y(t)$ . Suppose  $x(t)$  is bandlimited to  $|f| \leq B$ .

- (a) State the condition on  $f_s$  that guarantees  $y(t) = x(t)$ .

**Answer:**

$$f_s > 2B.$$

- (b) For  $B = 4$  kHz, determine the minimum theoretical sampling rate.

**Answer:**

$$f_s > 8 \text{ kHz}.$$

Thus 8 kHz is the limiting theoretical value; the strict theorem requires a rate greater than 8 kHz.

- (c) If  $f_s = 6$  kHz, describe qualitatively what happens to frequency components between 3 and 4 kHz.

**Answer:** Components from 3–4 kHz alias into 3–2 kHz.

- (d) Explain why ideal reconstruction cannot undo aliasing that has already occurred during sampling.

**Answer:** Aliasing is irreversible because multiple analogue frequencies have already mapped to the same discrete-time frequency.

## 9. Spectral interpretation of sampling

A continuous-time cosine of frequency 600 Hz is sampled at  $f_s = 2000$  samples/s.

- (a) Determine its normalized cyclic and normalized radian frequencies.

**Answer:**

$$f_0 = 600/2000 = 0.3 \text{ cycles/sample},$$

$$\hat{\omega}_0 = 0.6\pi \text{ rad/sample}.$$

- (b) Draw the two spectral lines inside the principal normalized-frequency interval.

**Answer:**

$$\hat{\omega} = \pm 0.6\pi.$$

**(c)** Show the nearest repeated spectral lines outside the principal interval.**Answer:**

$$\hat{\omega} = \pm 1.4\pi$$

are the nearest repeated lines outside the principal interval.

**(d)** Convert the repeated normalized-frequency locations back to analogue-frequency aliases.**Answer:** The repeated lines at  $\pm 1.4\pi$  correspond to analogue aliases at  $\pm 1400$  Hz.**(e)** Explain which analogue-frequency representative ideal reconstruction selects.**Answer:** The  $\pm 600$ -Hz representative inside  $|f| \leq 1000$  Hz.

## 10. Stroboscopic aliasing

A disk rotates clockwise at 750 revolutions/minute and is observed only when a strobe flashes.

**(a)** If the strobe flashes at 750 flashes/minute, describe the apparent motion of the disk.**Answer:** Stationary.**(b)** Repeat for 375 flashes/minute and explain why the disk can again appear stationary.**Answer:** Stationary (2 revolutions per flash).**(c)** For 806 flashes/minute, determine the angular advance or retreat observed from one flash to the next.**Answer:**

$$750/806 = 0.93052 \text{ rev/flash.}$$

$$0.93052 - 1 = -0.06948 \text{ rev/flash} \approx -25.0^\circ/\text{flash.}$$

Thus the observed motion is a small rotation in the opposite direction.

**(d)** Determine the apparent rotation rate in revolutions/minute for the 806 flashes/minute case.**Answer:**

$$f_{\text{app}} = 750 - 806 = -56 \text{ rev/min.}$$

So it appears to rotate at 56 rpm in the direction opposite to the true rotation.

**(e)** Relate this visual effect to aliasing of a sampled sinusoid.**Answer:** Rotational speeds differing by integer multiples of the flash rate are aliases.

## 11. Reconstructing a discrete-time sinusoid

A digital-to-continuous converter operates at  $f_s = 4000$  samples/s and receives  $x[n] = 3\cos(0.7\pi n - \pi/4)$ .**(a)** Determine the analogue frequency selected by ideal reconstruction inside the Nyquist interval.**Answer:**

$$f = (0.7\pi/2\pi)(4000) = 1400 \text{ Hz.}$$

**(b)** Write the reconstructed continuous-time sinusoid.**Answer:**

$$x_r(t) = 3 \cos(2\pi \cdot 1400t - \pi/4).$$

**(c)** List two other positive analogue frequencies that would produce the same sample sequence before reconstruction.

**Answer:** Two examples are 5400 Hz with phase  $-\pi/4$  and 2600 Hz with phase  $+\pi/4$ .

**(d)** What is the largest analogue frequency that this converter can reconstruct without ambiguity?

**Answer:**

$$f_N = f_s/2 = 2000 \text{ Hz.}$$

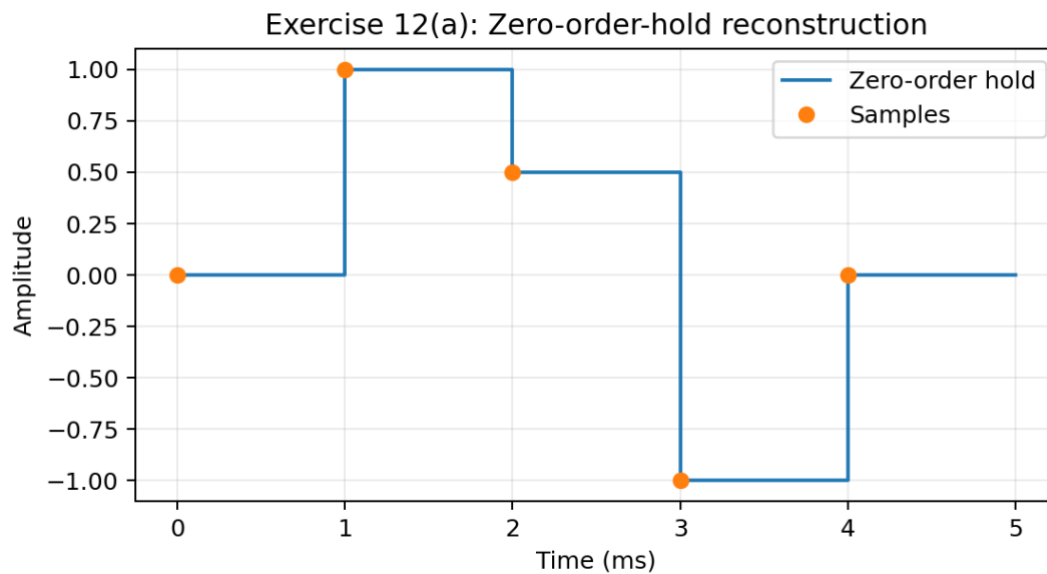
## 12. Sample-and-hold and linear interpolation

A sequence has sample values  $x[0] = 0$ ,  $x[1] = 1$ ,  $x[2] = 0.5$ ,  $x[3] = -1$ ,  $x[4] = 0$ .

The sampling period is  $T_s = 1 \text{ ms}$ .

**(a)** Sketch the zero-order-hold reconstruction from  $t = 0$  to 5 ms.

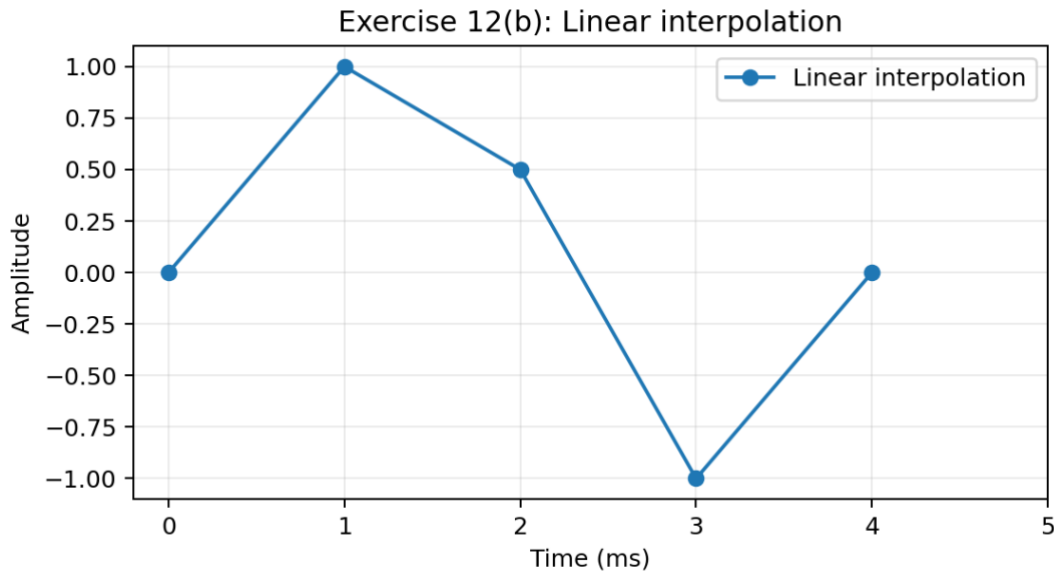
**Answer:** A zero-order hold keeps each sample value constant until the next sample.



*Zero-order-hold reconstruction.*

**(b)** Sketch the linear interpolation over the same interval.

**Answer:** Linear interpolation joins adjacent samples by straight-line segments.



*Linear interpolation between the available samples.*

(c) Compare the continuity and smoothness of the two reconstructions.

**Answer:** ZOH: staircase with jumps. Linear interpolation: continuous piecewise-linear curve.

(d) Explain why neither method is generally identical to ideal bandlimited interpolation.

**Answer:** Neither uses ideal sinc interpolation.

(e) State one practical reason why zero-order hold is commonly used in digital-to-analogue conversion.

**Answer:** Simple, causal, inexpensive hardware implementation.

### 13. Interpolation as a sum of shifted pulses

A reconstructed signal is formed from samples  $x[n]$  using an interpolation pulse  $p(t)$ :  $\hat{x}(t) = \sum_{n=-\infty}^{\infty} x[n]p(t - nT_s)$ .

(a) Explain the role of the shift  $nT_s$  and the multiplier  $x[n]$ .

**Answer:**  $nT_s$  positions the pulse;  $x[n]$  sets its amplitude.

(b) Describe the pulse  $p(t)$  for a zero-order hold.

**Answer:** Rectangular pulse of duration  $T_s$ .

(c) Describe qualitatively how a triangular pulse produces linear interpolation.

**Answer:** Overlapping triangular pulses produce straight lines between samples.

(d) What property must an ideal interpolation pulse have at the sample instants so that the reconstructed signal passes exactly through all sample values?

**Answer:**

$$p(kT_s) = \delta[k].$$

That is, the pulse must equal 1 at its own sample instant and 0 at all other sample instants (the cardinal interpolation property).

(e) Explain why the pulse shape affects frequencies introduced or attenuated by the reconstruction stage.

**Answer:** The pulse frequency response shapes the reconstructed spectrum.



## 14. Sampling a chirp

Consider a chirp whose instantaneous frequency increases linearly from 0 Hz at  $t = 0$  to 3000 Hz at  $t = 1$  s. The signal is sampled at  $f_s = 4000$  samples/s.

(a) Write the instantaneous frequency  $f_i(t)$  of the continuous-time chirp.

**Answer:**

$$f_i(t) = 3000t \text{ Hz}, \quad 0 \leq t \leq 1.$$

(b) Determine the time at which the chirp first reaches the Nyquist frequency.

**Answer:**

$$f_N = 4000/2 = 2000 \text{ Hz},$$

$$3000t = 2000 \Rightarrow t = 2/3 \text{ s} \approx 0.667 \text{ s}.$$

(c) For times after this instant, determine the apparent reconstructed frequency inside  $0 \leq f \leq f_s/2$ .

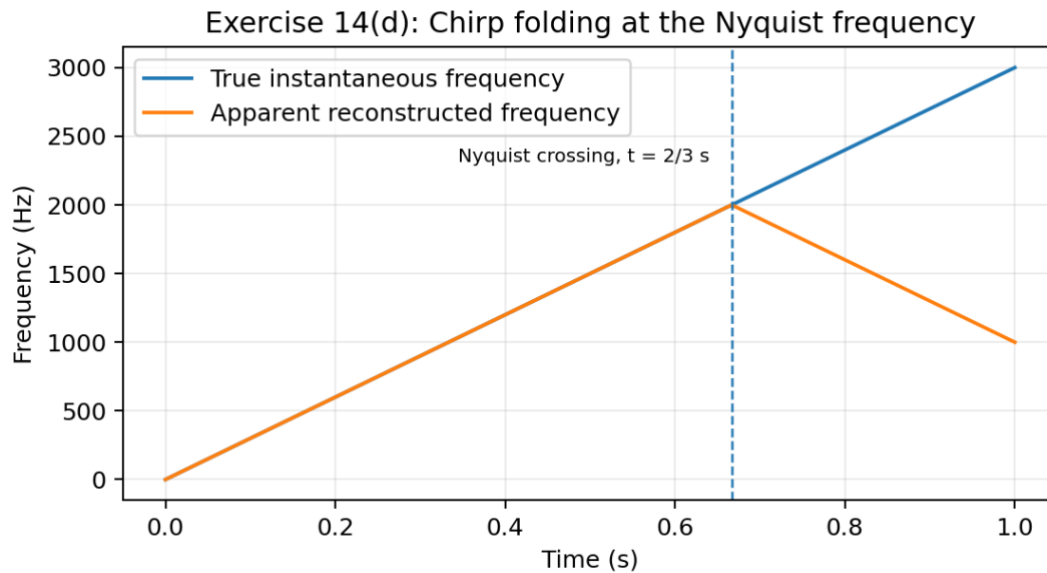
**Answer:**

$$f_{app}(t) = 4000 - 3000t, \quad 2/3 \leq t \leq 1.$$

It falls from 2000 Hz at  $t=2/3$  s to 1000 Hz at  $t=1$  s.

(d) Sketch the true instantaneous frequency and the aliased apparent frequency versus time from 0 to 1 s.

**Answer:** The true frequency continues to rise, while the reconstructed alias folds downward after the Nyquist crossing.



*True and aliased instantaneous frequencies.*

(e) Describe what the corresponding sampled chirp would sound or look like near and beyond the folding point.

**Answer:** The apparent frequency rises to 2 kHz, then folds downward to 1 kHz.

## 15. Integrated computational investigation

Use Python with NumPy and Matplotlib to investigate sampling and reconstruction.

- (a) Generate a 100-Hz continuous-time cosine on a dense time grid and sample it at 1000, 250, 180, and 120 samples/s.

**Answer:**

$$x[n] = \cos(2\pi \cdot 100n/f_s).$$

- (b) For each sampling rate, overlay the sample points on the original waveform and identify whether aliasing occurs.

**Answer:** No aliasing: 1000, 250 samples/s. Aliasing: 180→80 Hz; 120→20 Hz.

- (c) For each case, calculate the normalized radian frequency and its principal-interval equivalent.

**Answer:**

$$1000: \hat{\omega} = 0.2\pi,$$

$$250: \hat{\omega} = 0.8\pi,$$

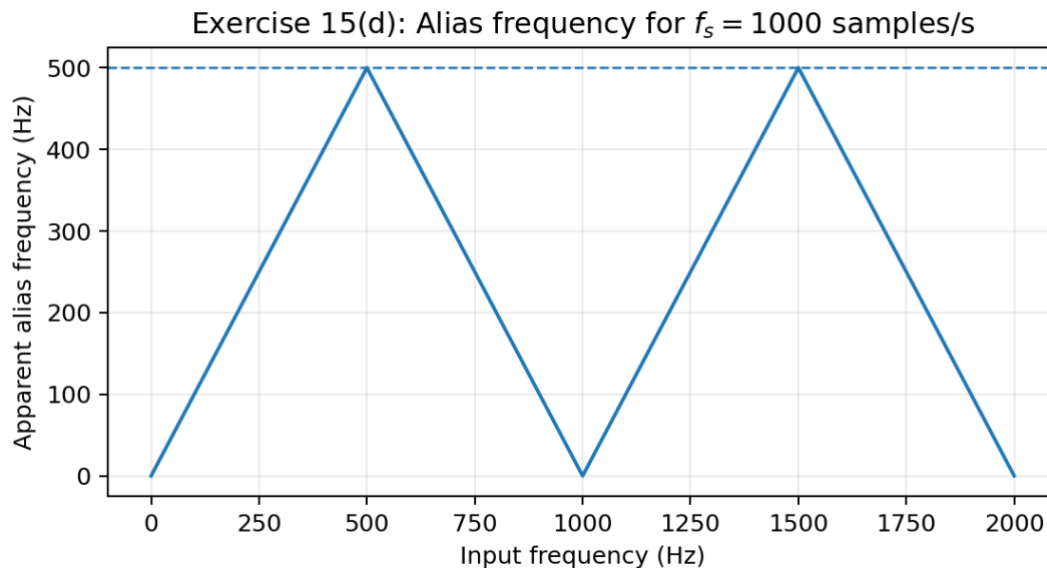
$$180: \hat{\omega} = 1.111 \dots \pi \equiv -0.888 \dots \pi,$$

$$120: \hat{\omega} = 1.666 \dots \pi \equiv -0.333 \dots \pi.$$

- (d) Plot the apparent reconstructed alias frequency as the input frequency is swept from 0 to  $2f_s$  for a fixed  $f_s$ .

**Answer:** For fixed  $f_s$ , the apparent baseband frequency is the folded distance to the nearest integer multiple of  $f_s$ :

$$f_{\{alias\}} = \min(r, f_s - r), \quad r = f \bmod f_s.$$



*Alias-frequency folding for  $f_s=1000$  samples/s.*

- (e) For one sample sequence, compare zero-order-hold and linear interpolation.

**Answer:** ZOH: staircase. Linear: straight-line interpolation.

- (f) Summarize how the time-domain plots, normalized-frequency calculations, and reconstruction plots illustrate the sampling theorem.

**Answer:** Above Nyquist the original baseband signal is uniquely reconstructible; below Nyquist, folding creates irreversible aliases.